
Intelligence Engineering

Solving Real World Problems At Scale

Prof. Dr.-Ing. Mark Schutera | September 25, 2026 · *drifting stream*

LECTURE IX

Continual Learning and Test in Production

This lecture in one slide

IX Continual Learning and Test in Production

Continual Learning

Test in Production

*why this lecture matters, and what
you can do after it*

1

Continual Learning

Stateless Retraining Versus Stateful Training

Stateless retraining

Stateful training

Why Continual Learning?

$P(y|x)$

[1]

Continual Learning Challenges

Fresh data access

Evaluation

Algorithm

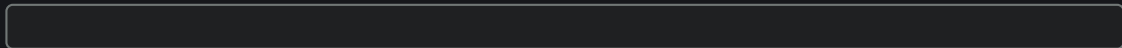
Fresh data access challenge

Fresh data access

[1]

$P(y|x)$

Evaluation challenge



Algorithm challenge

$P(\hat{y} | x)$

[1]

Four Stages of Continual Learning

Stage 1

Stage 2

Stage 3

Stage 4

Stage 1: Manual, stateless retraining

$P(\hat{y} | x)$

[1]

Stage 2: Automated retraining

Automated retraining



Stage 3: Automated, stateful training

$P(y|x)$

[1]

Stage 4: Continual learning

Definition: Continual learning

How Often to Update Your Models

$P(\hat{y} | x)$

[1]

Value of data freshness

Data freshness

Model iteration versus data iteration

Model iteration

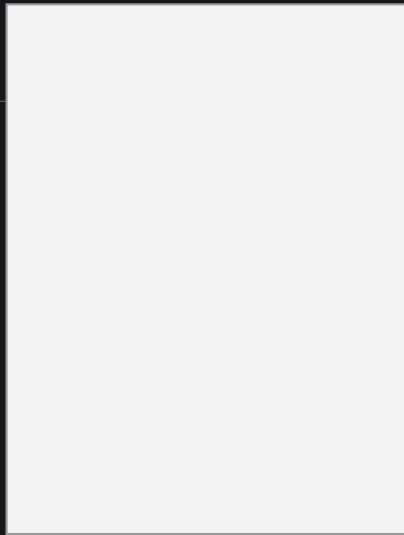
Data iteration

2

Test in Production

Shadow Deployment

Shadow deployment



$P(y | x)$

[1]

A/B Testing

$$P(y|x)$$

$$[1]$$

Canary Release

Definition: Canary release

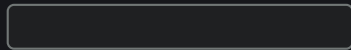
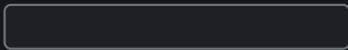
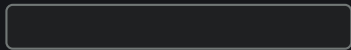
Interleaving Experiments

$P(y|x)$

[1]

Bandits

$$P(\hat{y} | x)$$



$$[1]$$

Contextual bandits as an exploration strategy

$P(y|x)$

Definition: Contextual bandits

[1]

3

Summary

Summary

$P(y|x)$

[1]

References

- 1 Chip Huyen. *Designing Machine Learning Systems: An Iterative Process for Production-Ready Applications*. Sebastopol, CA: O'Reilly Media, 2022.